

SPIDER

447 series
218 patients
4 hospitals

expert masks:
vertebrae,
discs, canal

vendor, field
(DICOM)

R1, R2 intensity

signal inside masks:
raw (R1); ratios,
z scores (R2)

R3 mask geometry

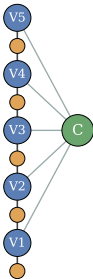
volumes, spacing,
length, counts,
voxel volume (mm)

R4 graph-relational

Laplacian spectrum
of spine graph +
scale-free ratios

spine graph

V vertebra, D disc,
C canal hub



Scanner prediction

R1 to R4, random
forest, patient
grouped CV;
AUC, bootstrap,
permutation

Controls

voxel size,
coverage,
sequence,
fixed field,
case mix, ComBat

Utility

degeneration,
narrowing;
within domain
and across
vendors

Series network

k-NN graph:
assortativity,
communities,
patient null